

D

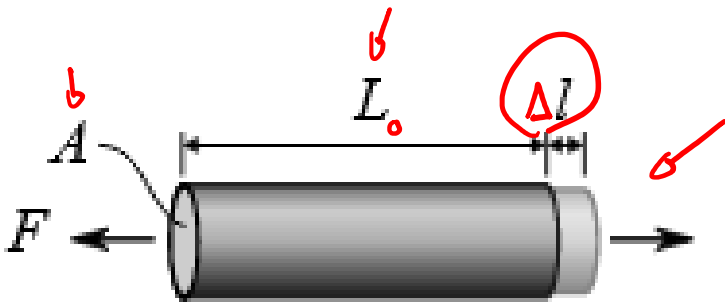
STM - Exercise 1

Stefano De La Pierre

Exercise 1

Use figure 1 to:

- a) define which test and which class of materials is referred to and why;
- b) calculate elastic modulus,
- c) Calculate the yield strength (0.2 % offset) and the tensile strength,
- d) Calculate the elongation at fracture for a 50 mm long specimen (write on curves data you use).
- e) Calculate the length of the specimen when you keep it loaded in A.



$$\sigma = \frac{F}{A}$$

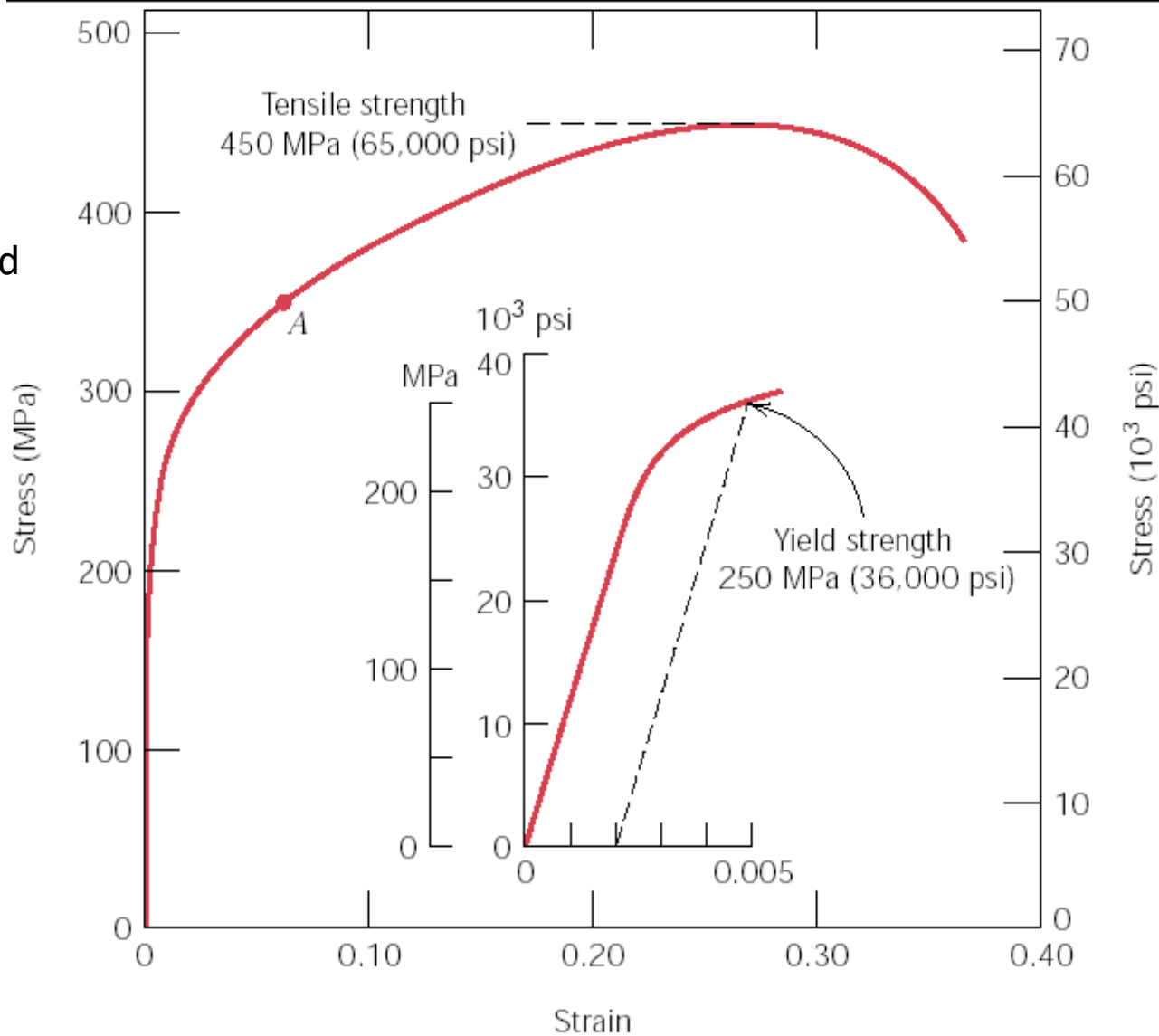
where

- σ stress [Pa]
- F applied force [N]
- A cross-sectional area [m²]

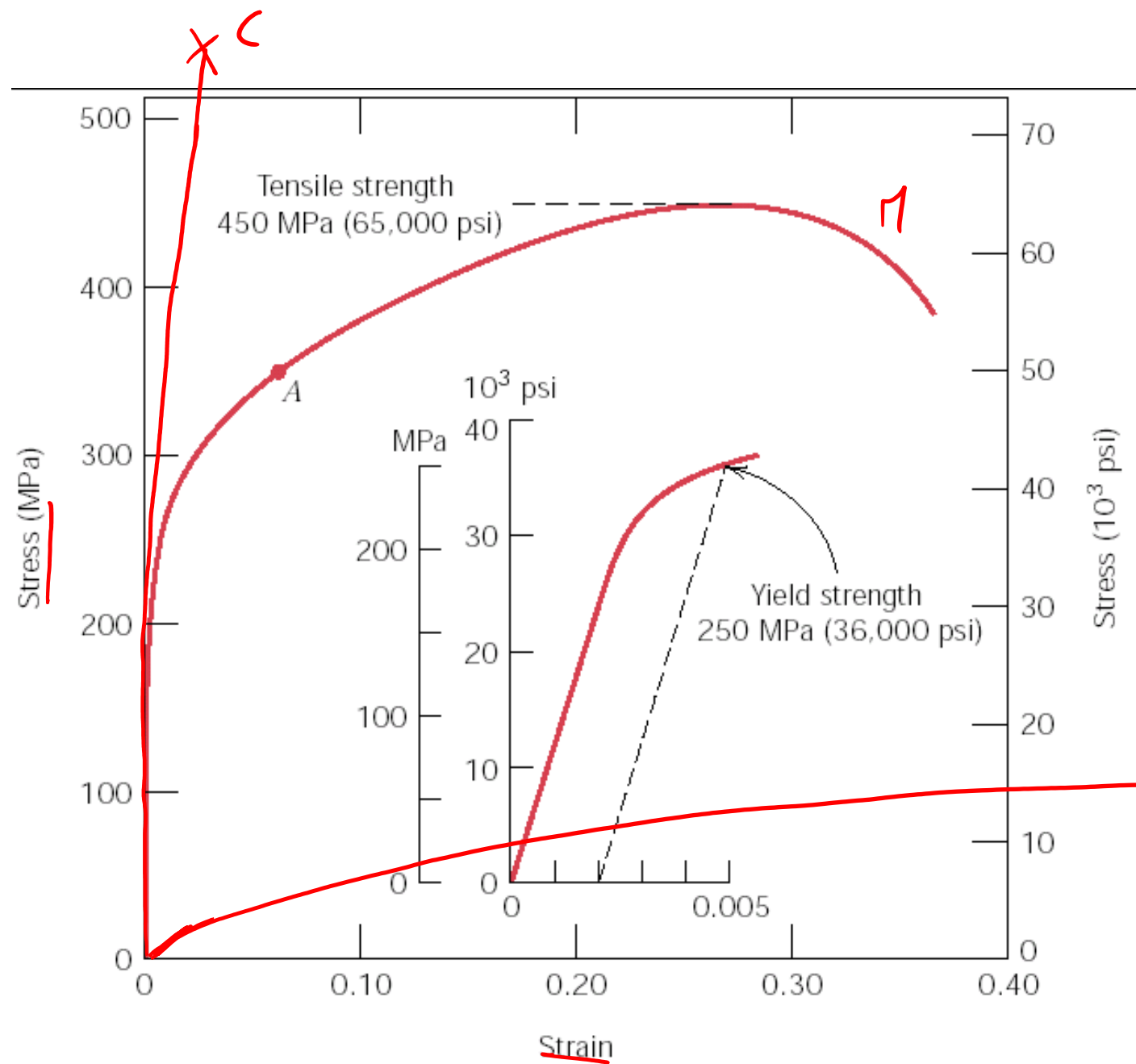
$$\epsilon = \frac{\Delta L}{L_0}$$

where

- ϵ strain
- ΔL total elongation [m]
- L_0 original length [m]



Exercise 1a: Use figure 1 to define which test and which class of materials is referred to and why

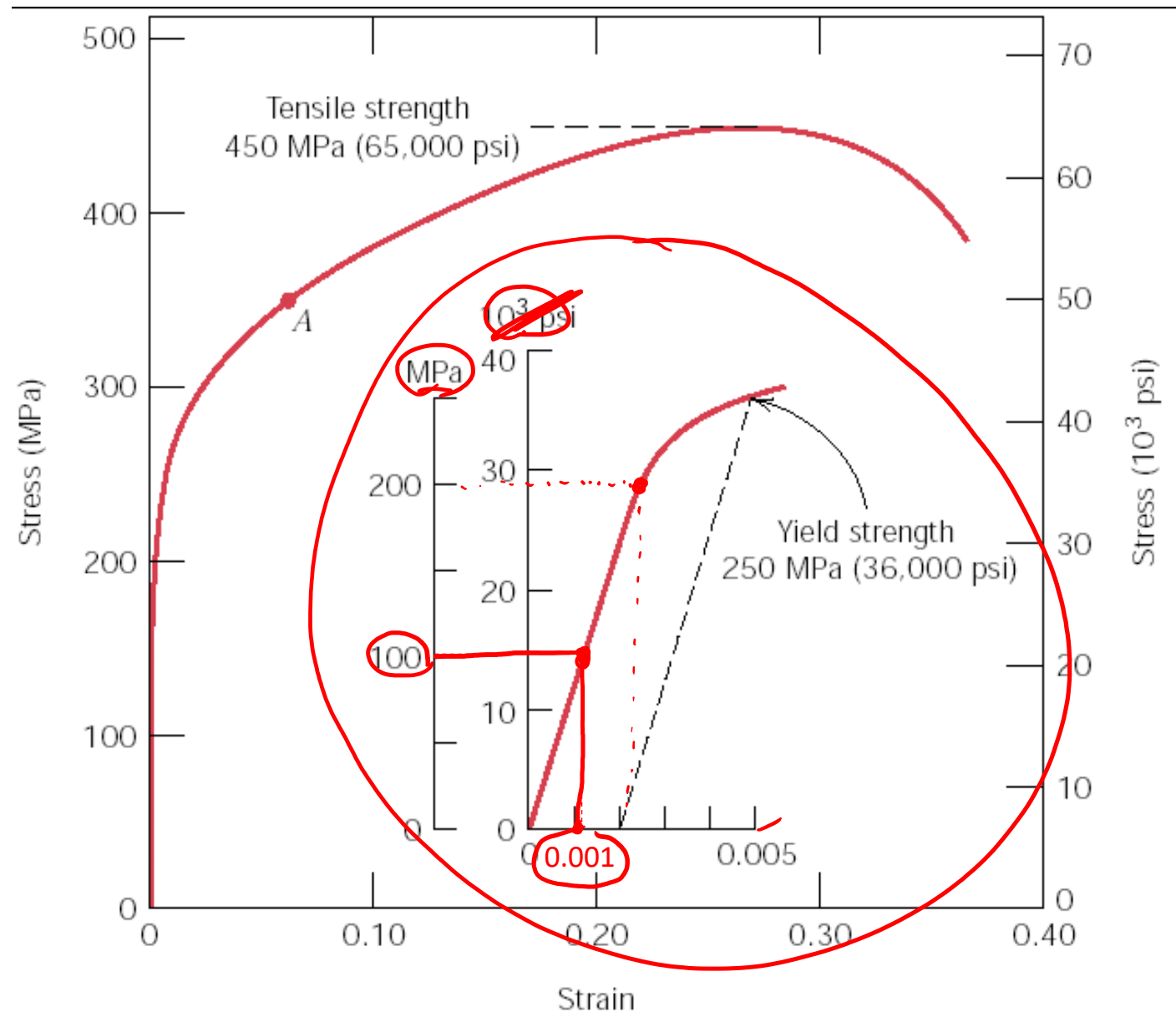


TENSILE

METAL

Typical stress-strain curve from tensile test
Ductile behavior and stress-strain values typical of
metallic materials

Exercise 1b: Calculate elastic modulus

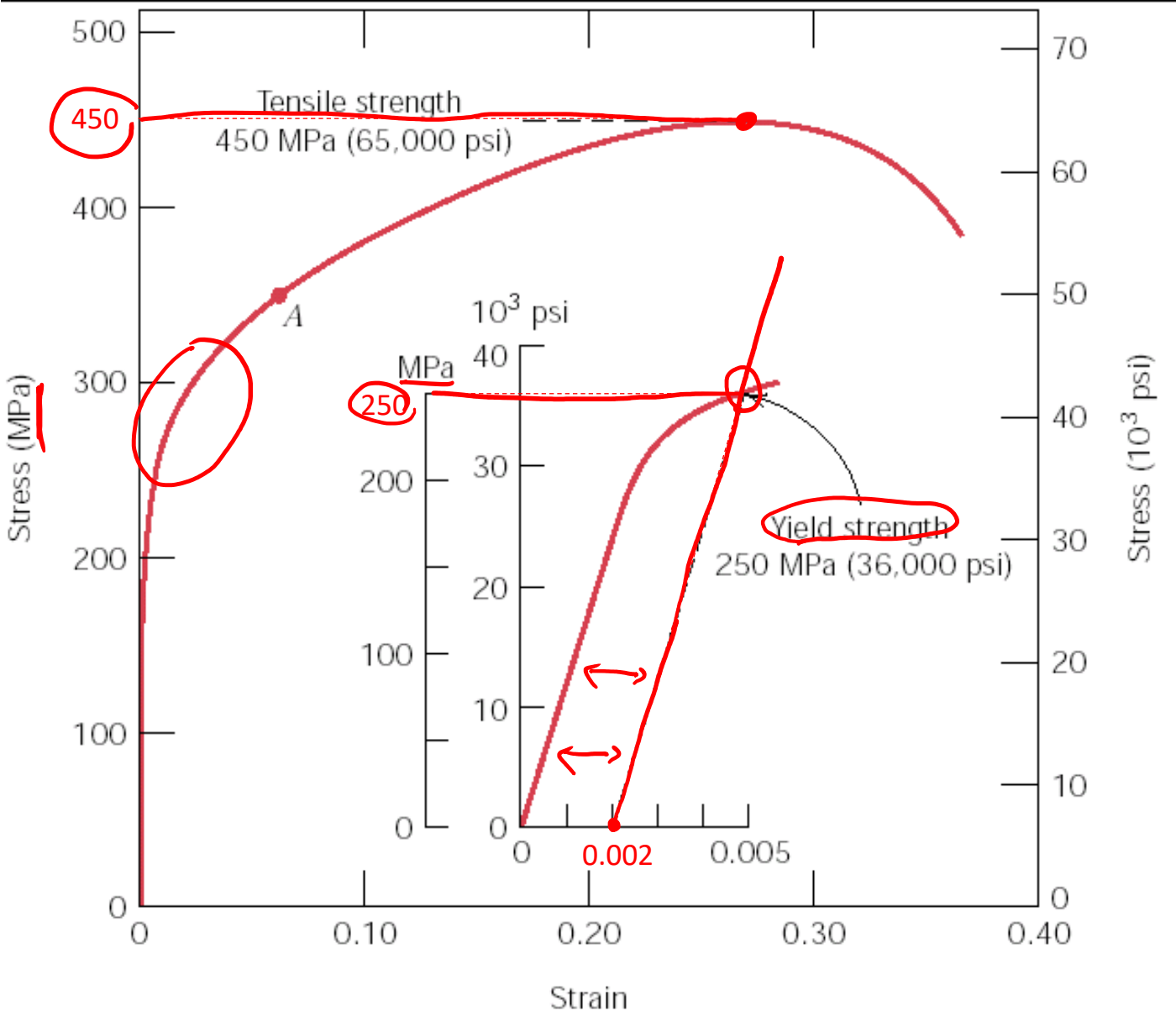


↓

$$E = \frac{\sigma}{\epsilon}$$

$$E = \frac{100 \text{ MPa}}{0,001} = 100.000 \text{ MPa} = 100 \text{ GPa}$$

Exercise 1c: calculate yield strength (0.2 % offset) and tensile strength



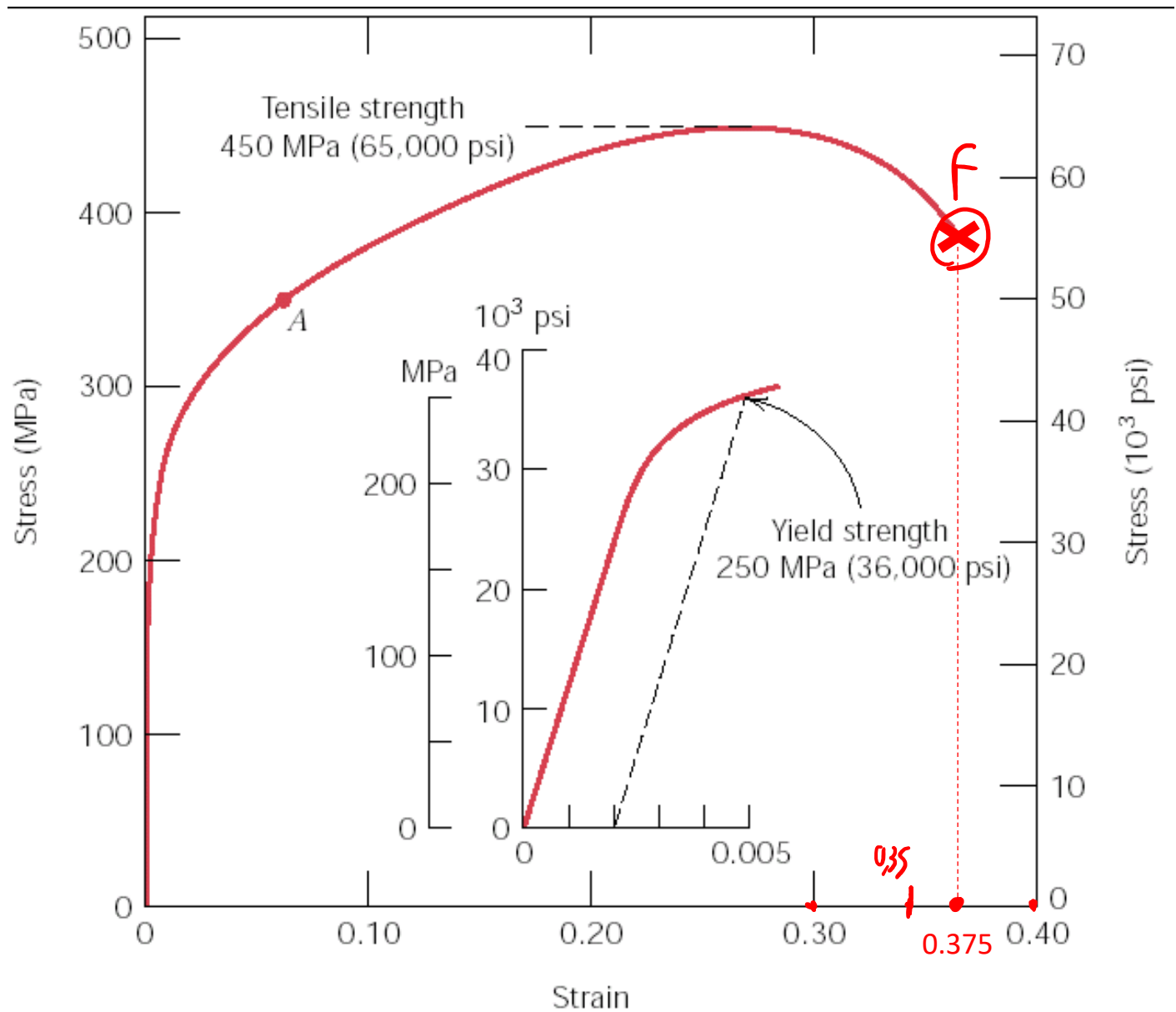
$$\epsilon = 0,2 \% = \underline{0,002}$$

$$\sigma_y = 250 \text{ MPa}$$

$$\sigma_t = \sigma_{\max} = \text{UTS}$$

$$\sigma_t = 450 \text{ MPa}$$

Exercise 1d: Calculate elongation at fracture for a 50 mm long specimen (write on curves data you use).



$\Delta l = ?$

$\epsilon_F = 0,375$

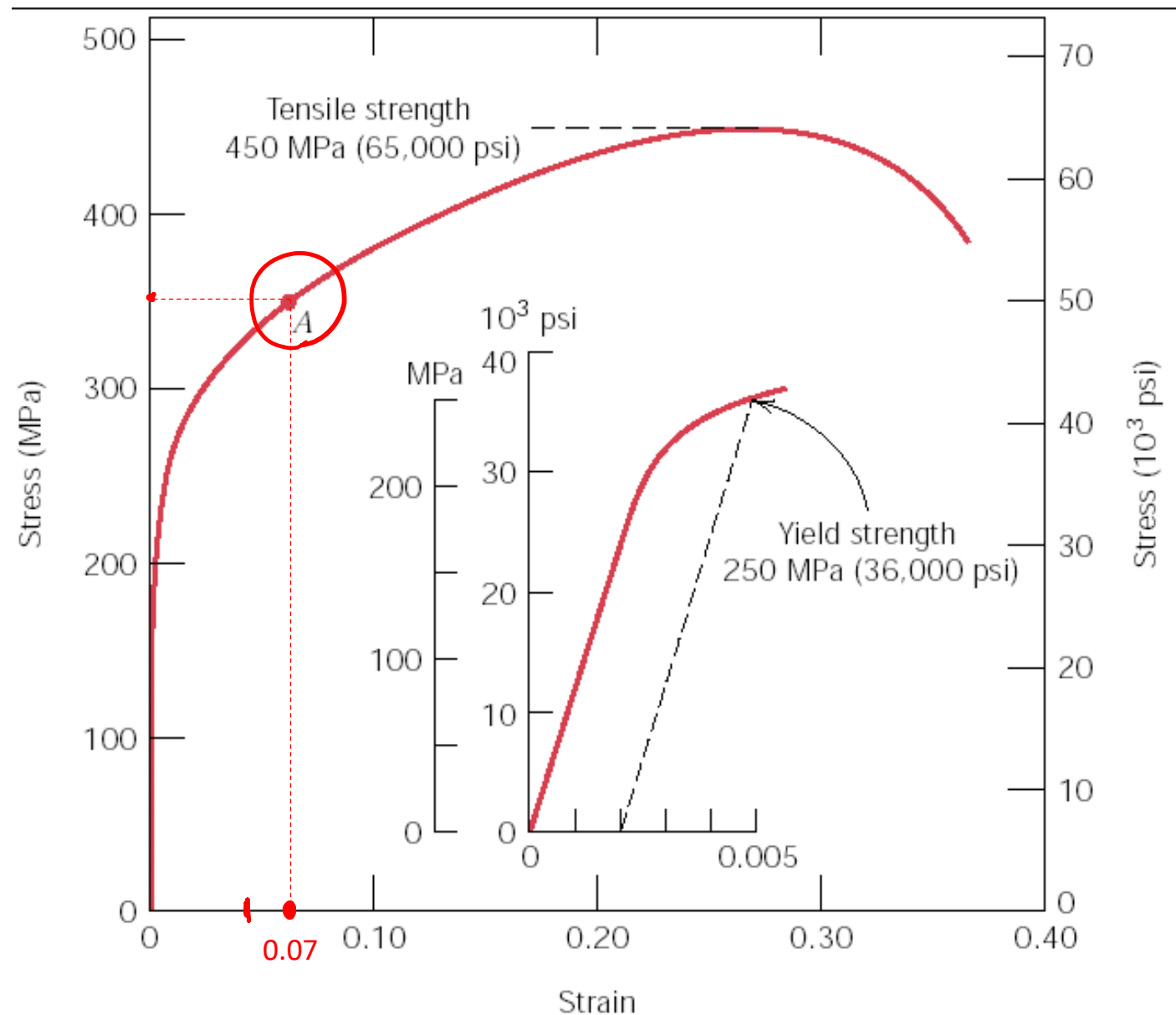
$\epsilon = \frac{\Delta l}{L_0} \rightarrow \Delta l = L_0 \cdot \epsilon$

Δl = 50 mm · 0,375 = 18,75 mm

Exercise 1e- Calculate the length of the specimen when you keep it loaded in (A).

$\uparrow$

$L_A = ?$



$$\epsilon_A = 0,07$$

$$\Delta L_A = \epsilon_A \cdot L_0 = 0,07 \cdot 50 \text{ mm} = 3,5 \text{ mm}$$

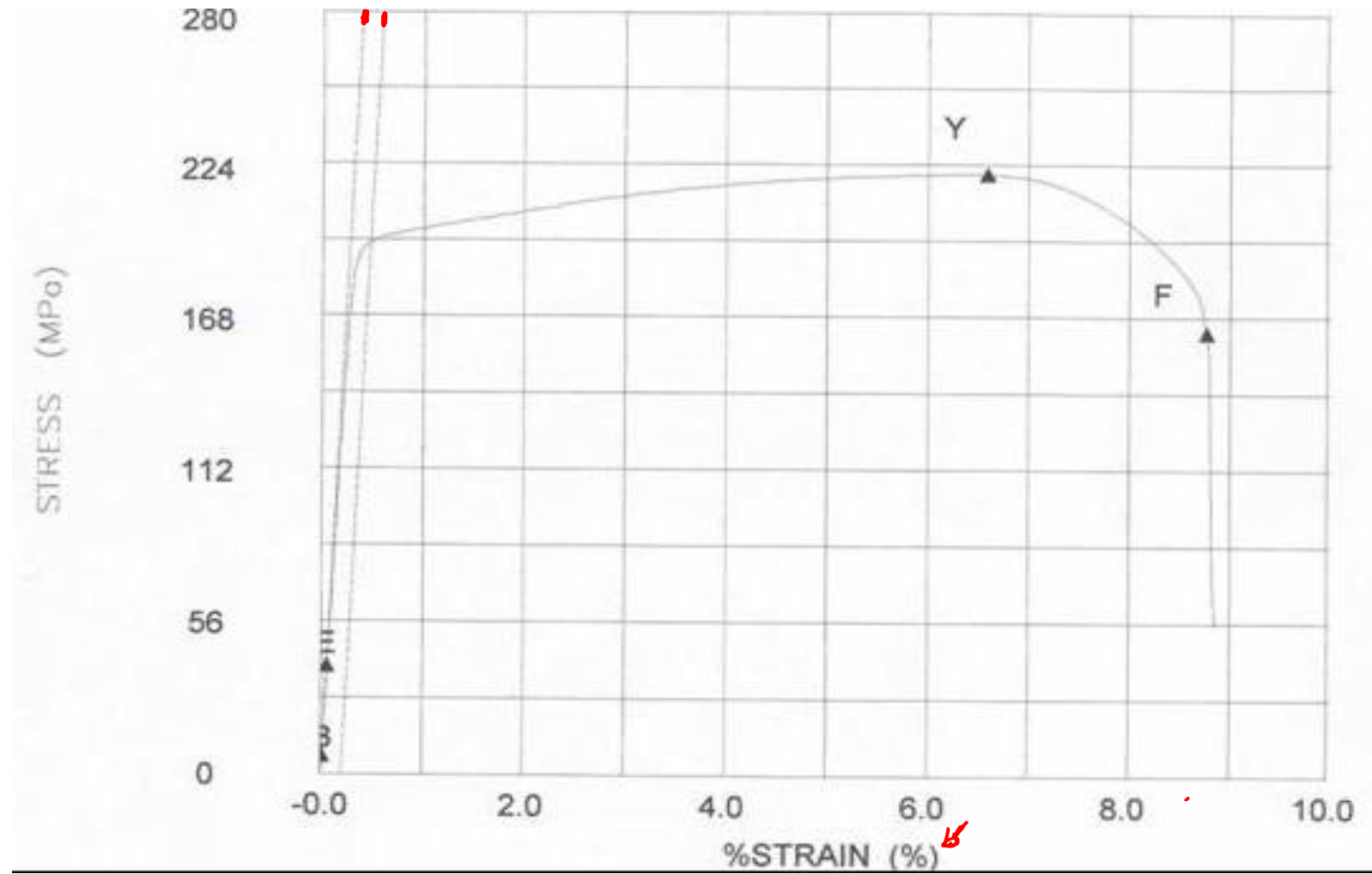
$$L_A = L_0 + \Delta L_A = 50 + 3,5 = 53,5 \text{ mm}$$

If you keep the sample loaded no recovery of elastic deformation occurs

Exercise 2: Use the graph below to define

- which test and which class of materials is referred to and why;
- to calculate elastic modulus,
- To calculate the yield strength (0,2 % strain) and the tensile strength,
- To calculate the elongation at fracture for a 50 mm long specimen;
- To calculate the elongation of this sample when loaded at 200 MPa, then unloaded.

(write on graph stress and strain
you use to calculate).



Exercise 2a: Define which test and which class of materials is referred to and why

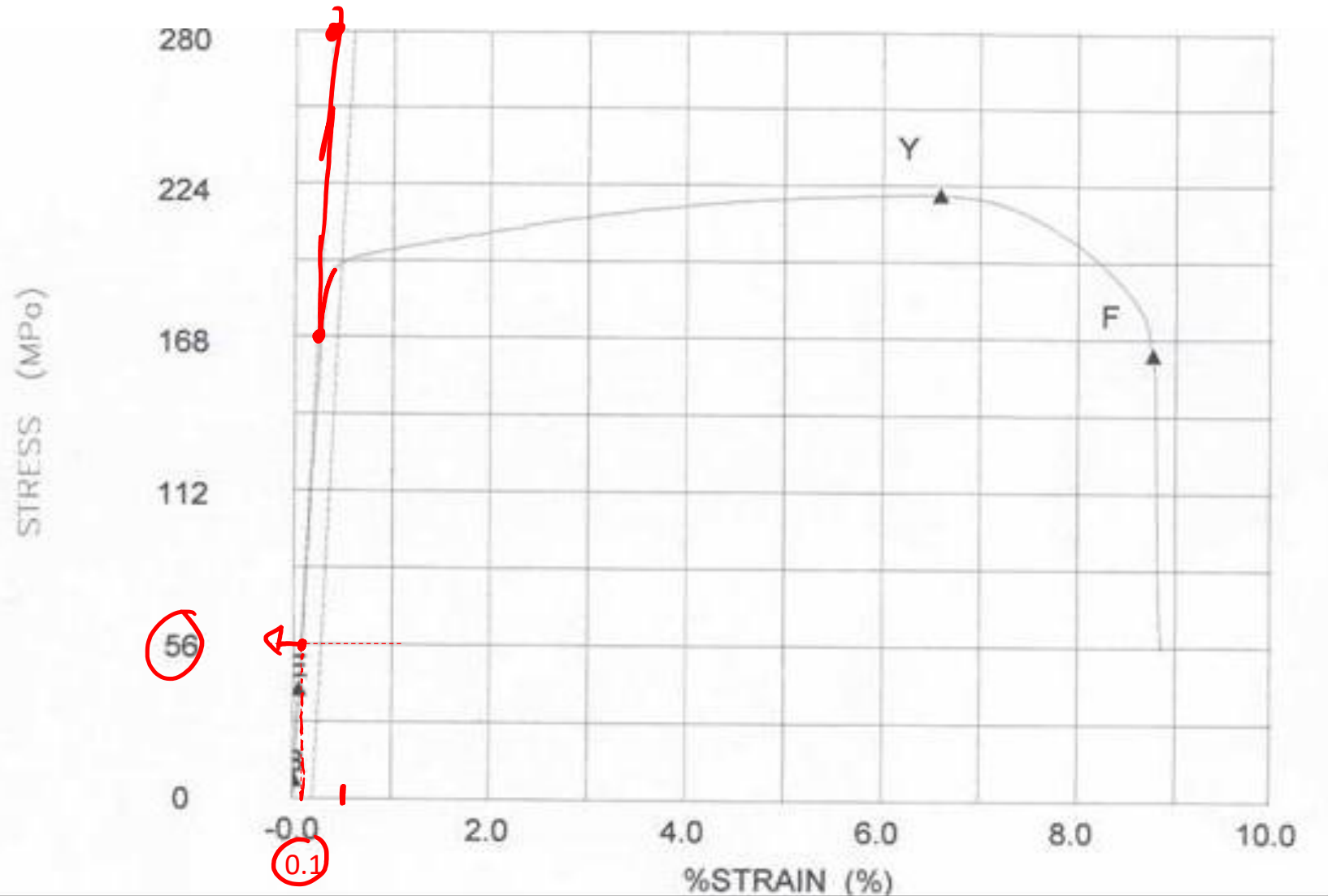


TENSILE

METALLIC

Typical stress-strain curve from tensile test
Ductile behavior and stress-strain values typical
of metallic materials

Exercise 2b: calculate elastic modulus



$$E = 42 \text{ GPa}$$

70 GPa ←

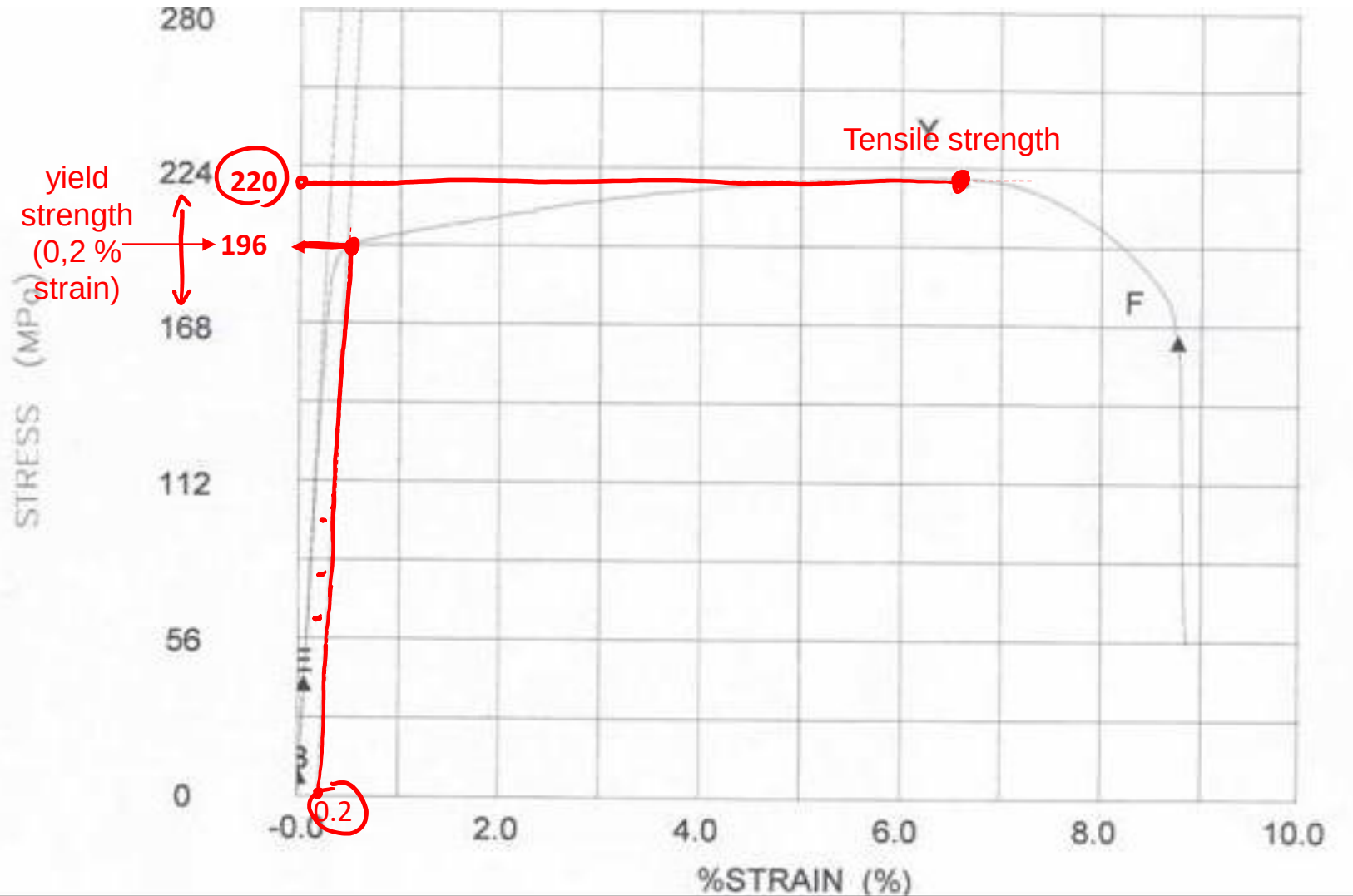
$$E = \frac{\sigma}{\epsilon} = \frac{56 \text{ MPa}}{0.001} = 56000 \text{ MPa}$$

↳ 56 GPa

$$\epsilon = 0.1\% = 0.001$$

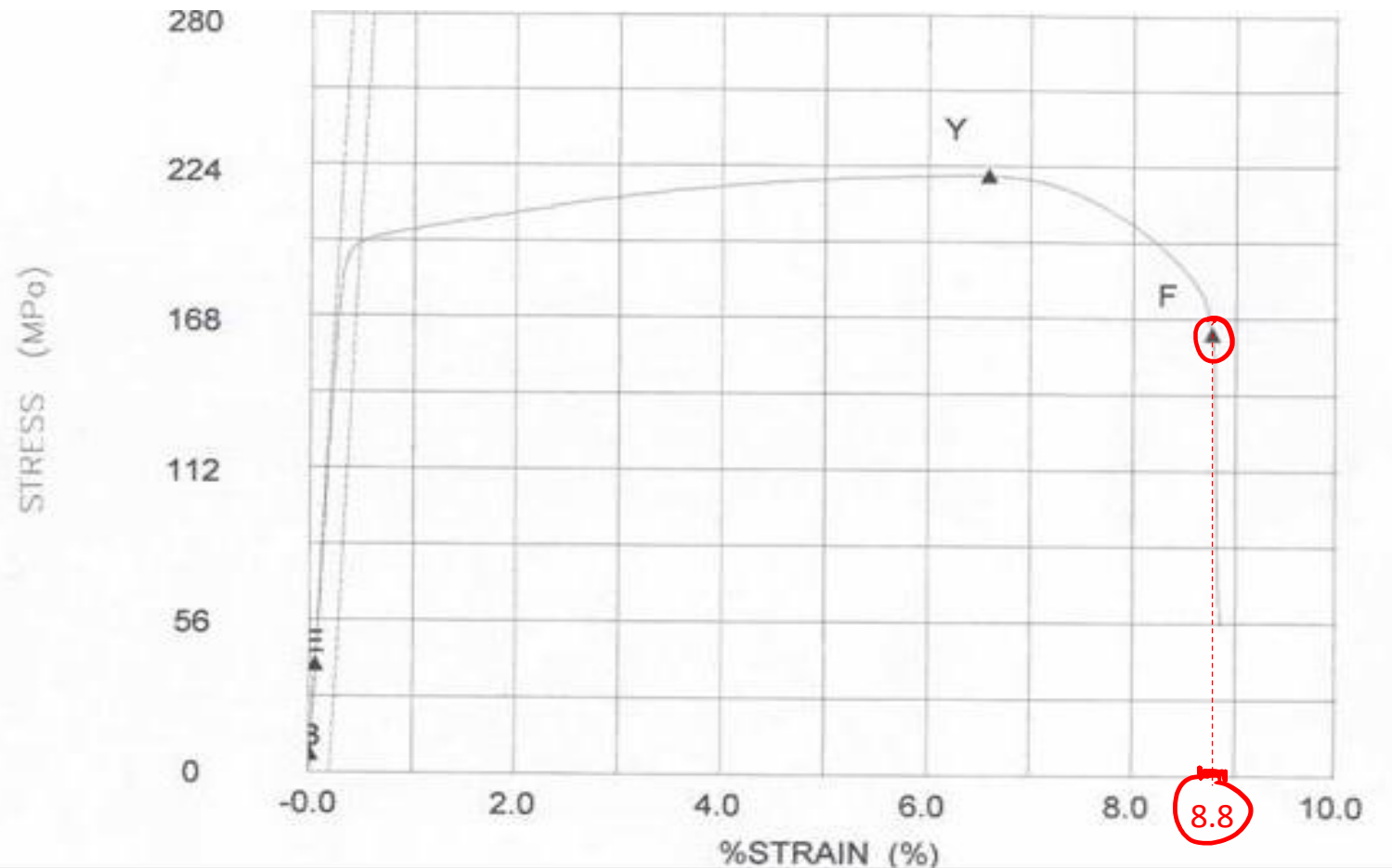
Exercise 2c: calculate yield strength (0,2 % strain) and tensile strength

$$\sigma_y = 196 \text{ MPa}$$



$$\sigma_t = 218 \text{ MPa}$$
$$= 220 \text{ MPa}$$

Exercise 2d: Calculate elongation at fracture for a 50 mm long specimen



$$\Delta l_F = \begin{cases} 4,35 \text{ mm} \\ 4,4 \text{ ''} \\ 4,5 \text{ ''} \end{cases}$$

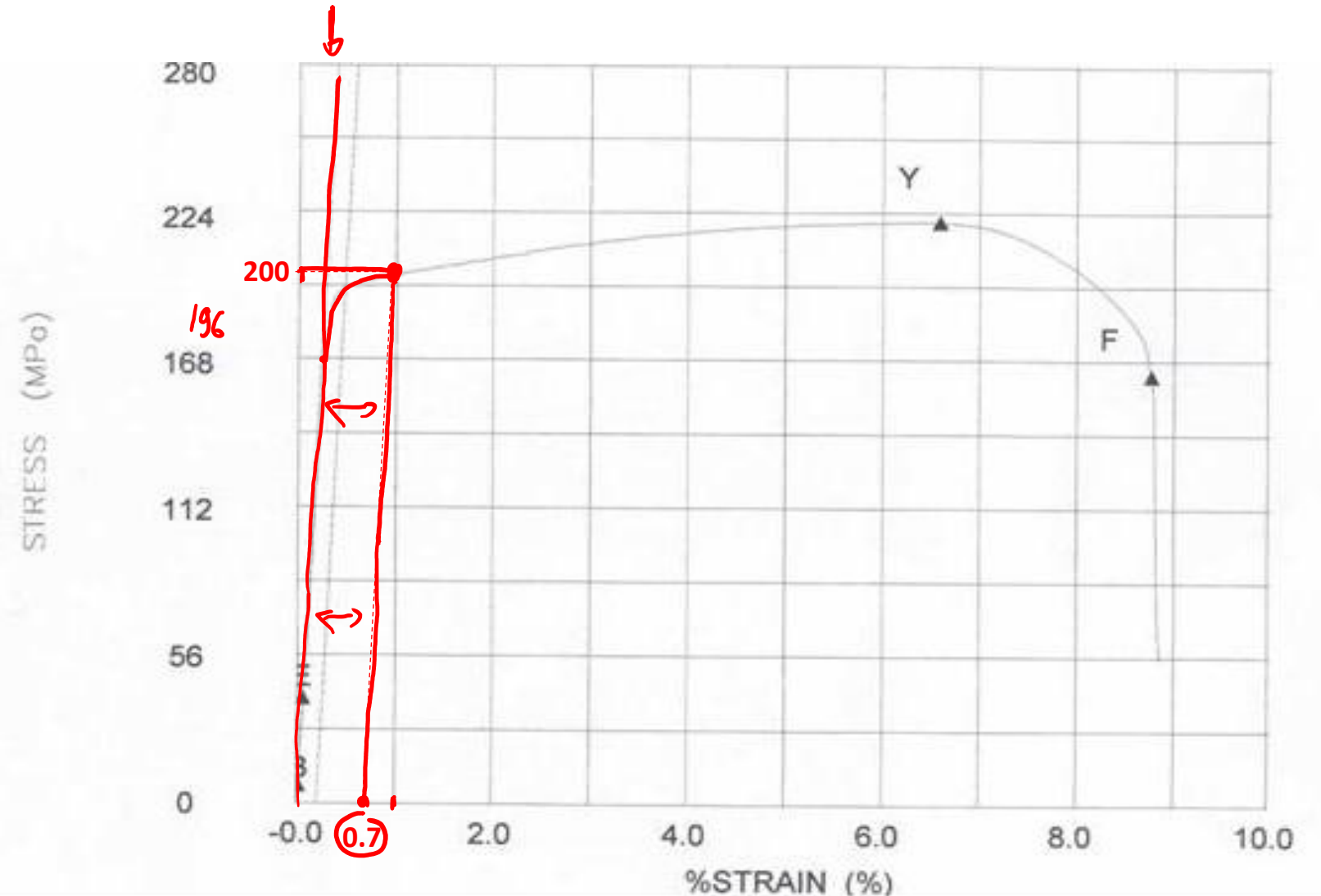
$$\epsilon = \frac{\Delta l}{L_0} \rightarrow \Delta l_F = \epsilon_F \cdot L_0 =$$

$$\Delta l_F = 0,088 \cdot 50 \text{ mm} = 4,4 \text{ mm}$$

$$\epsilon = 88\% = 0,88$$

Exercise 2e: Calculate the elongation of this sample when loaded at 200 MPa, then unloaded. (write on graph stress and strain you use to calculate).

$$\Delta l = 0,35 \text{ mm}$$



$$\Delta l = \varepsilon \cdot L_0$$

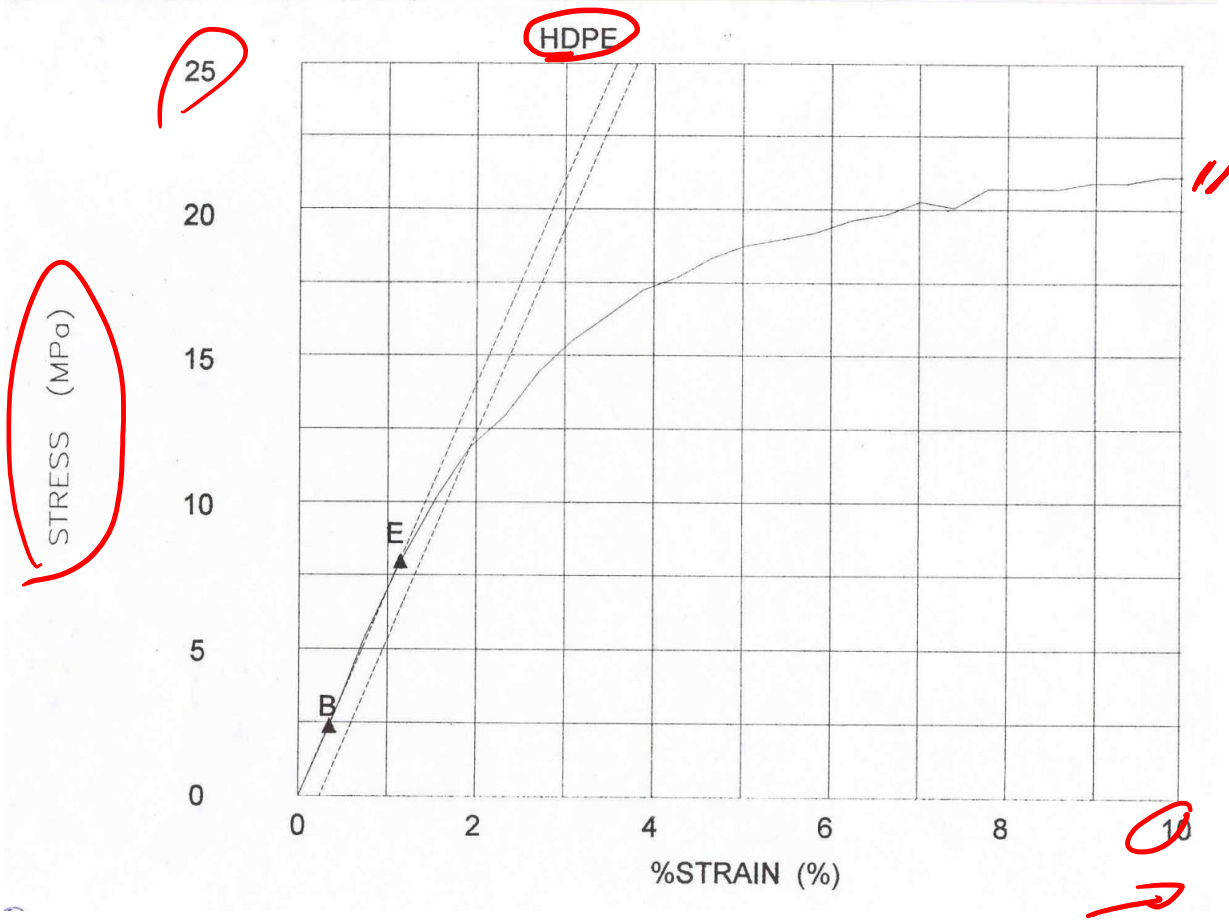
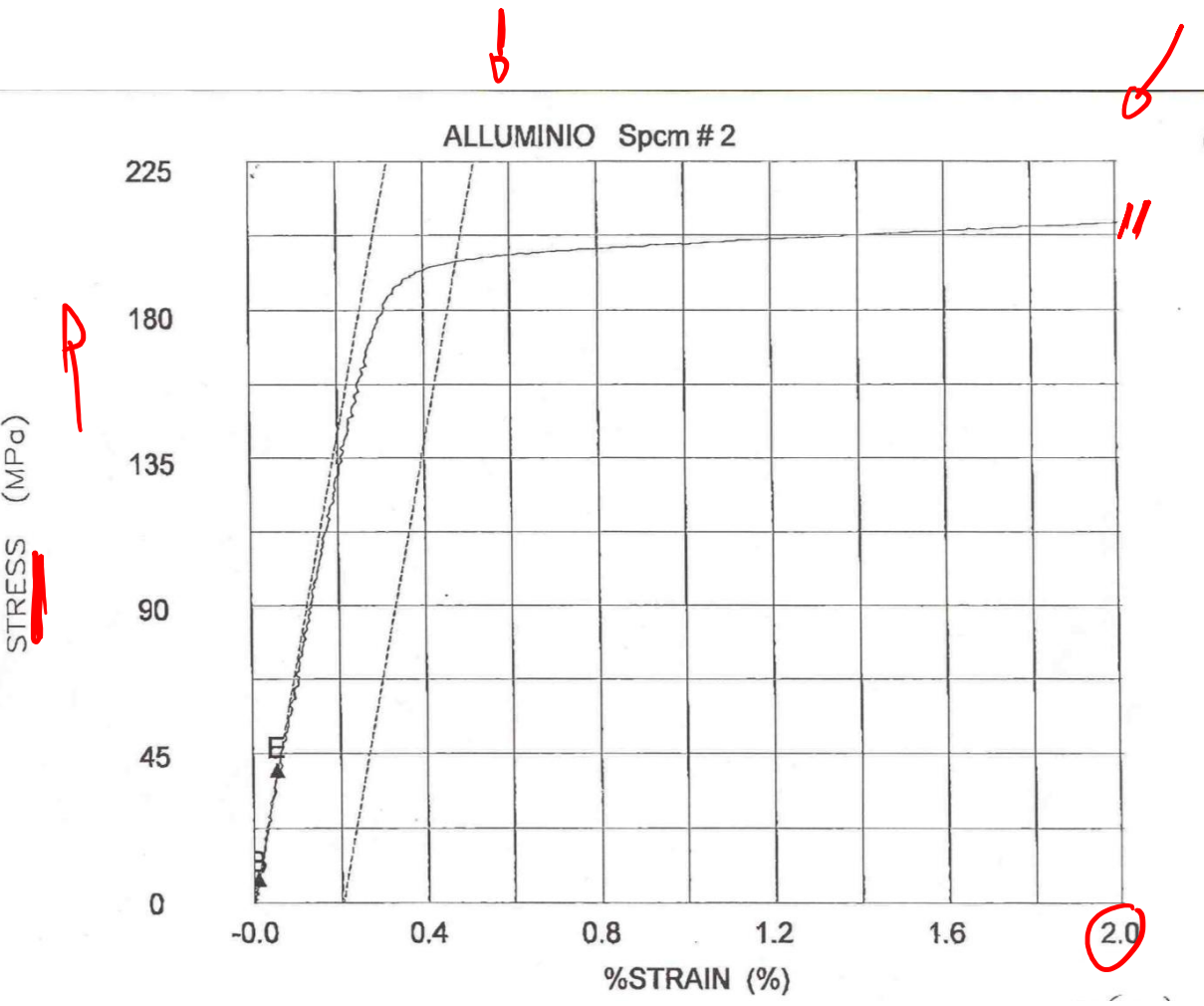
$$\varepsilon = 0,7\% = 0,007$$

$$\Delta l = 0,007 \cdot 50 \text{ mm} = 0,35 \text{ mm}$$

Loading and unloading →

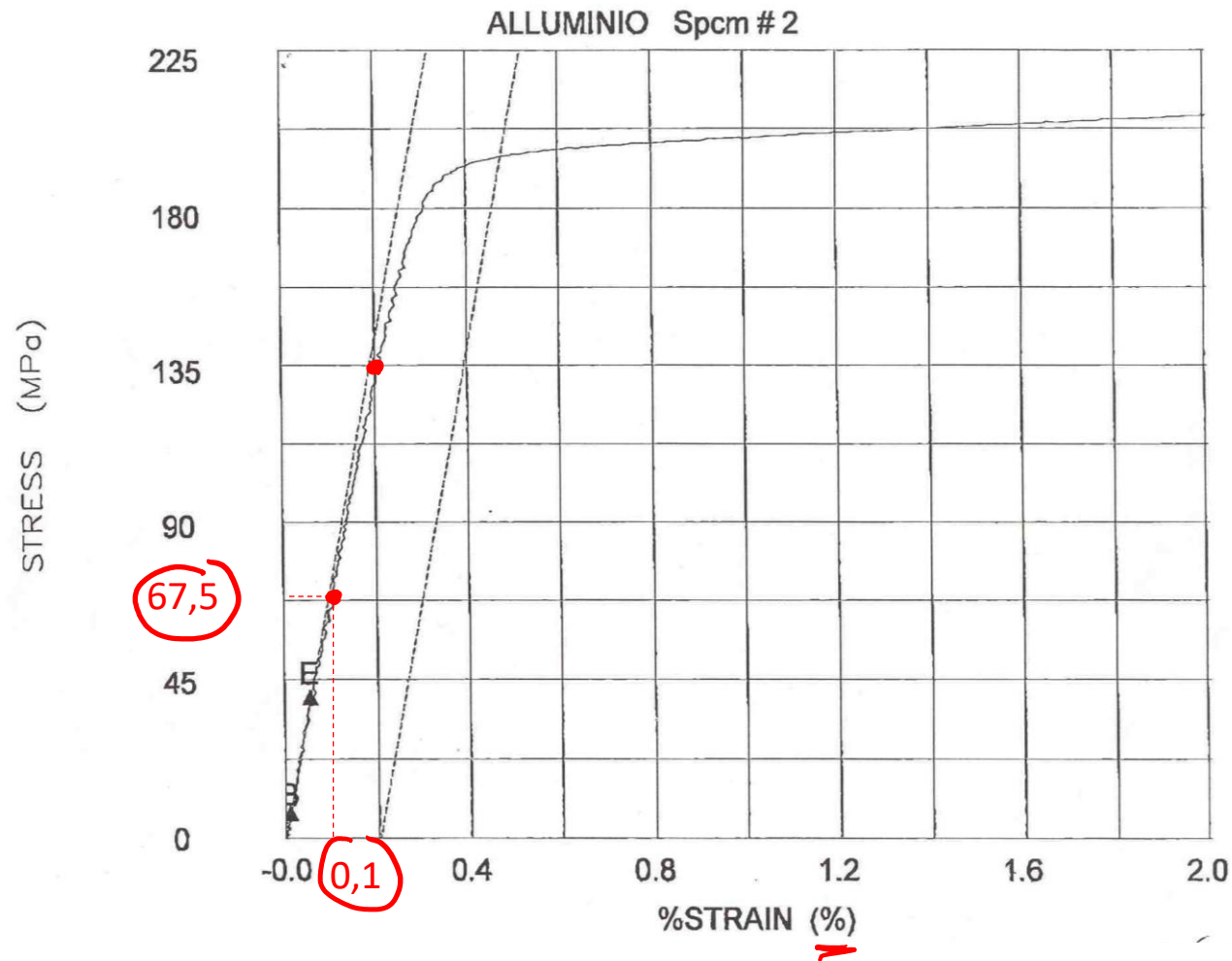
elastic deformation recovered,
only permanent deformation
present

Exercise 3: Determine the Elastic Modulus of Aluminum and High Density Polyethylene (HDPE) from the graphs below. Comment the obtained results.



Exercise 3 – case 1: Aluminum

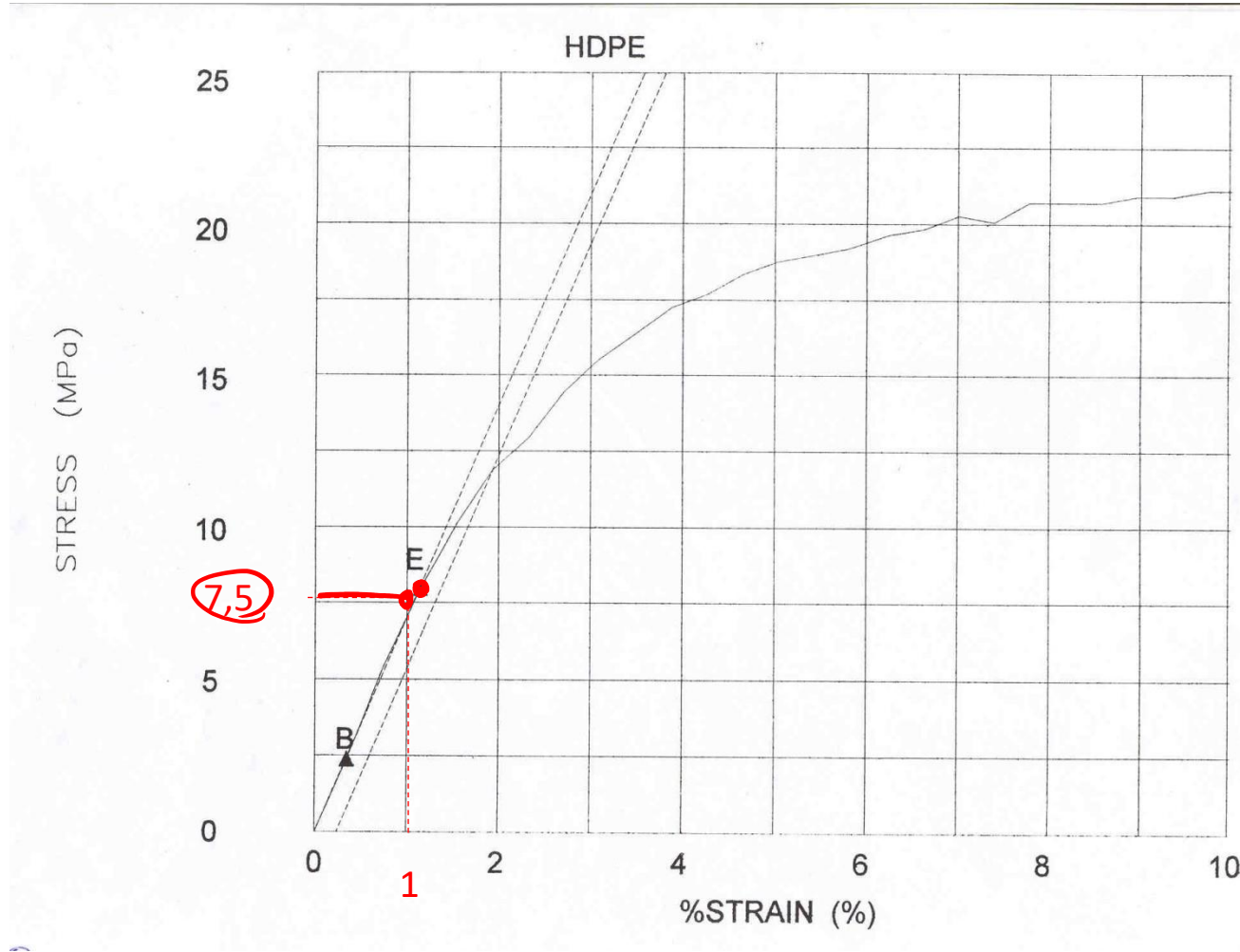
Elastic modulus (E) = slope of the linear (elastic) tract of the σ - ϵ curve



$$E = \frac{\sigma}{\epsilon} = \frac{67,5 \text{ MPa}}{0,001} = 67.500 \text{ MPa} \\ = 67,5 \text{ GPa}$$

Exercise 3 – case 2: HDPE

Elastic modulus (E) = slope of the linear (elastic) tract of the σ - ϵ curve



$$E = \frac{7,5 \text{ MPa}}{0,01} = 750 \text{ MPa}$$

0,75 GPa

Exercise 3 comparison Aluminum – HDPE

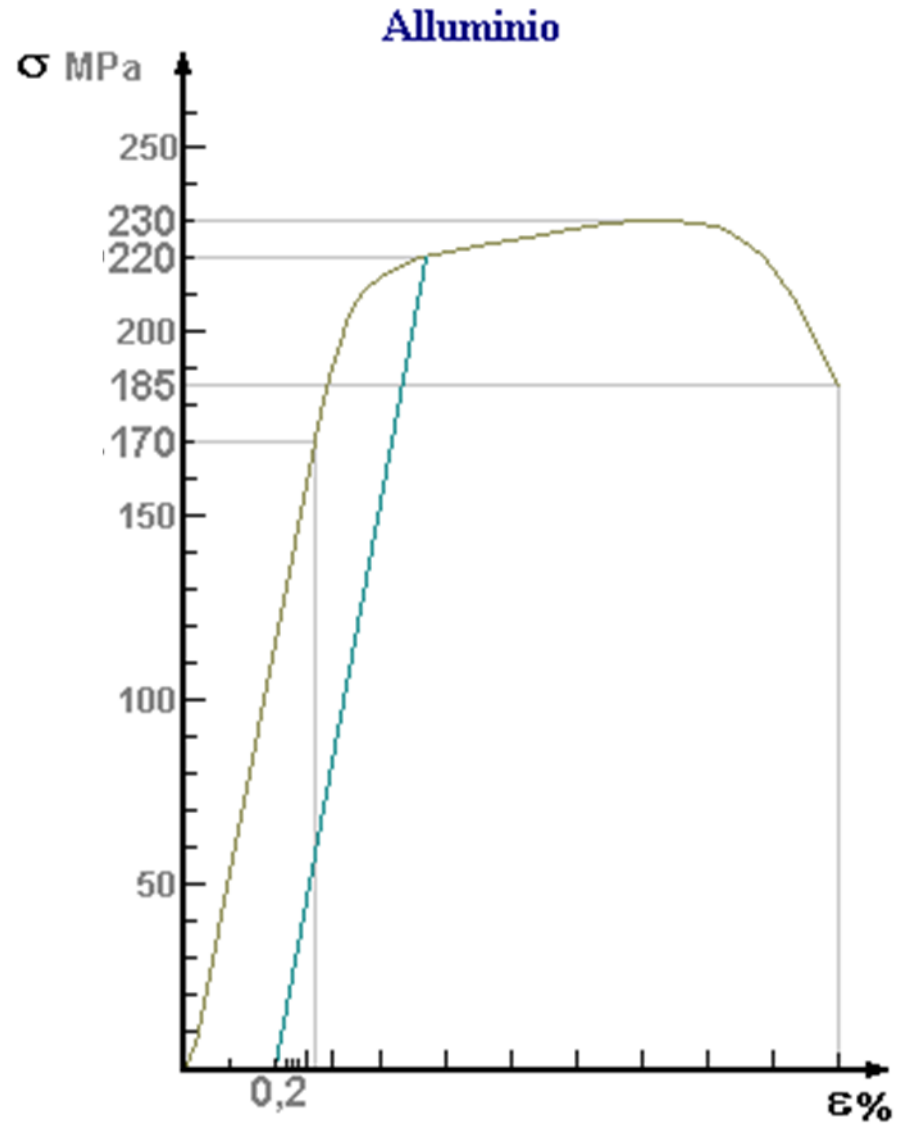
$E(\text{Al}) = 67.5 \text{ GPa}$

$E(\text{HDPE}) = 0.75 \text{ GPa}$ 

1 order of magnitude difference

Elastic modulus depends on the bonding strength: metallic bond vs van der Waals interactions

Quiz 1



Consider the σ - ϵ curve reported in the figure.

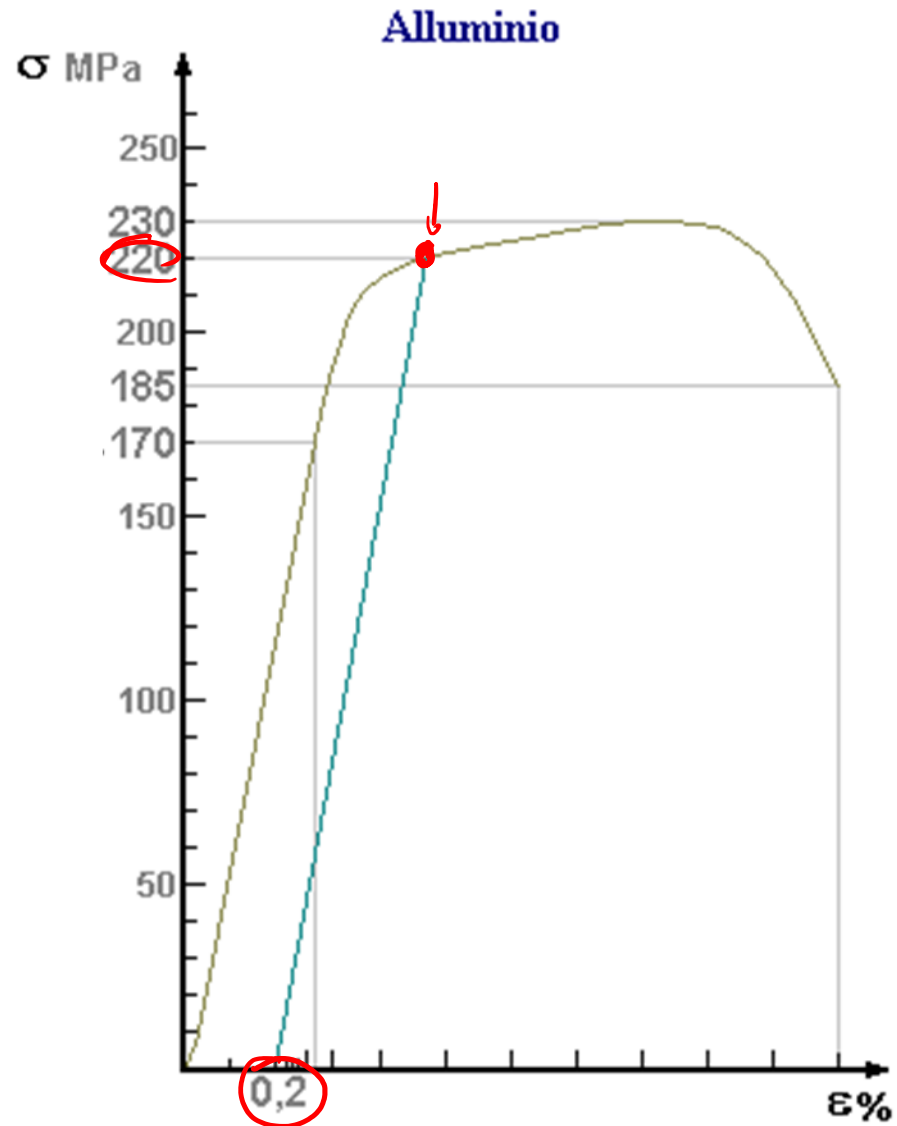
The yield strength (0.2 % offset) is:

a) 230 MPa

b) 220 MPa

c) 185 MPa

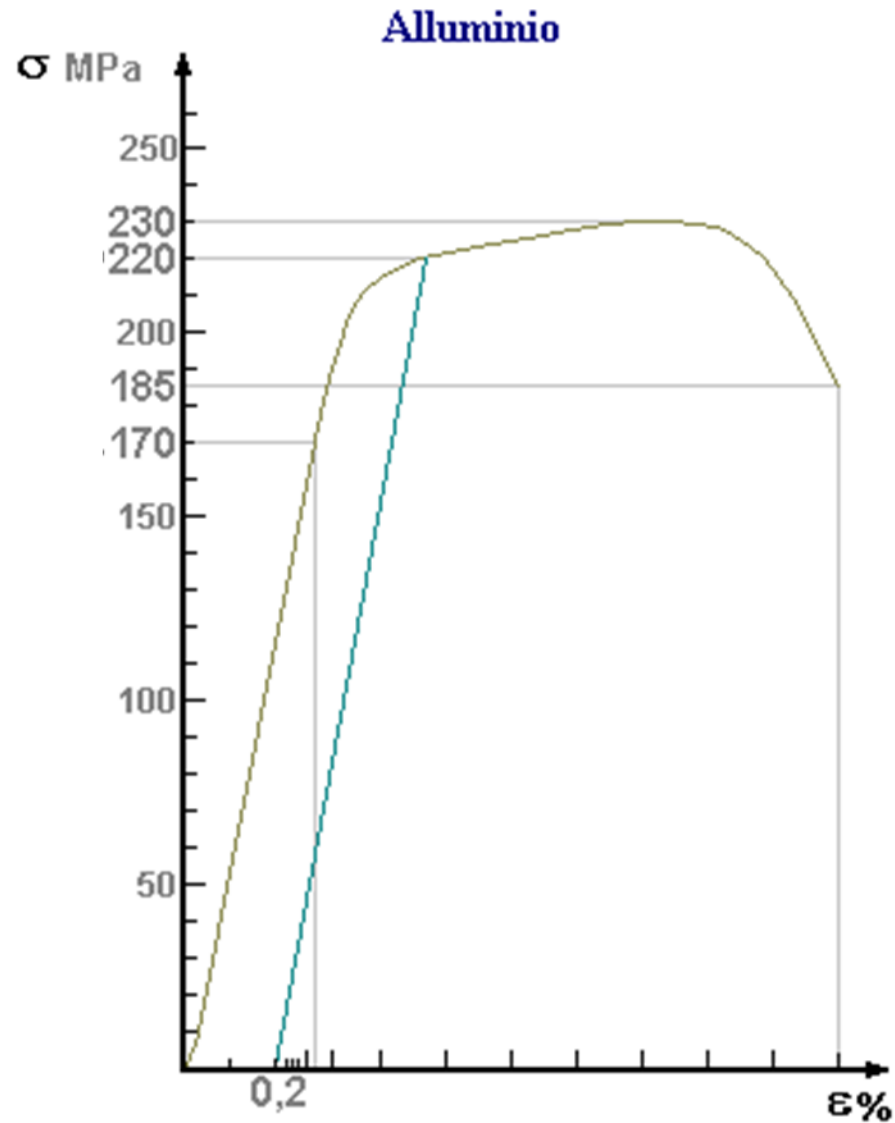
Quiz 1



Consider the σ - ϵ curve reported in the figure. The yield strength (0.2 % offset) is:

- a) 230 Mpa
- b) 220 Mpa**
- c) 185 MPa

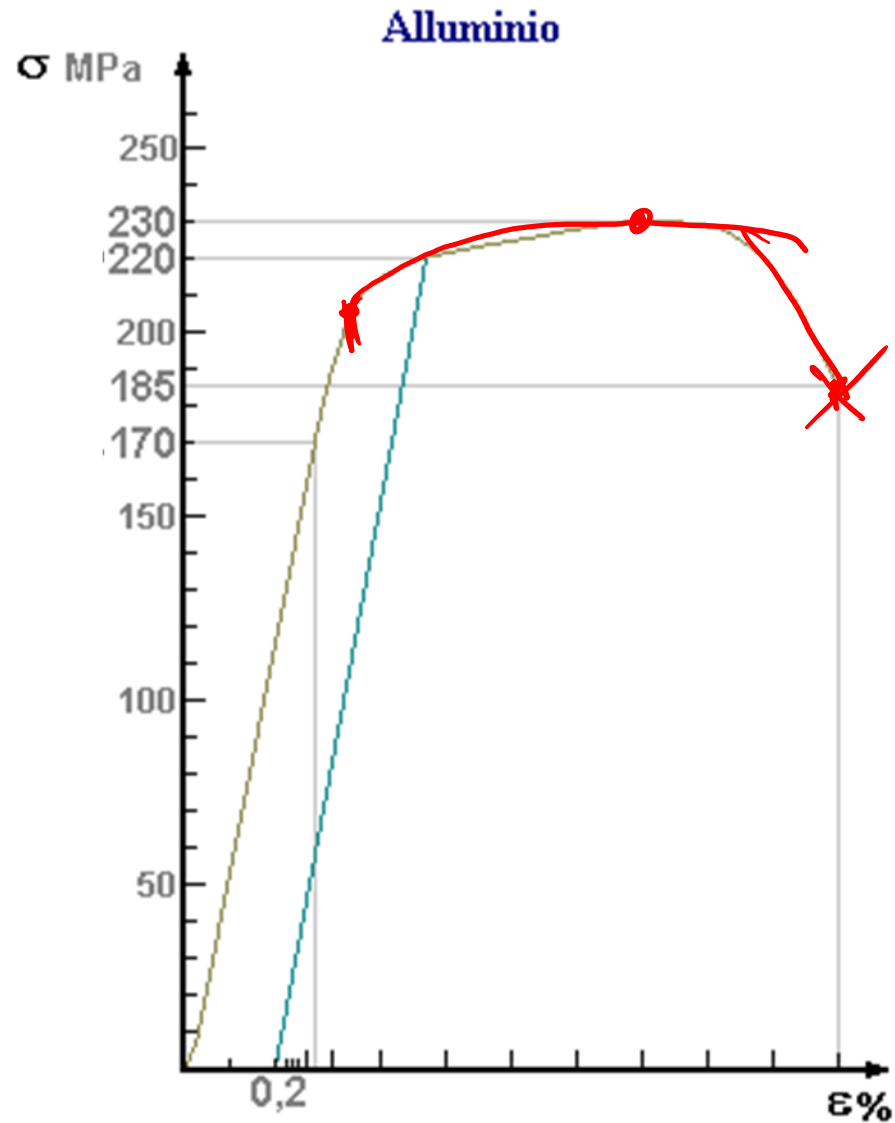
Quiz 2



The σ - ϵ curve in figure comes from the tensile test of an aluminum specimen. Select the right sentence (only one):

- a) The specimen underwent plastic deformation
- b) The specimen underwent fragile fracture
- c) The specimen did not have permanent deformations

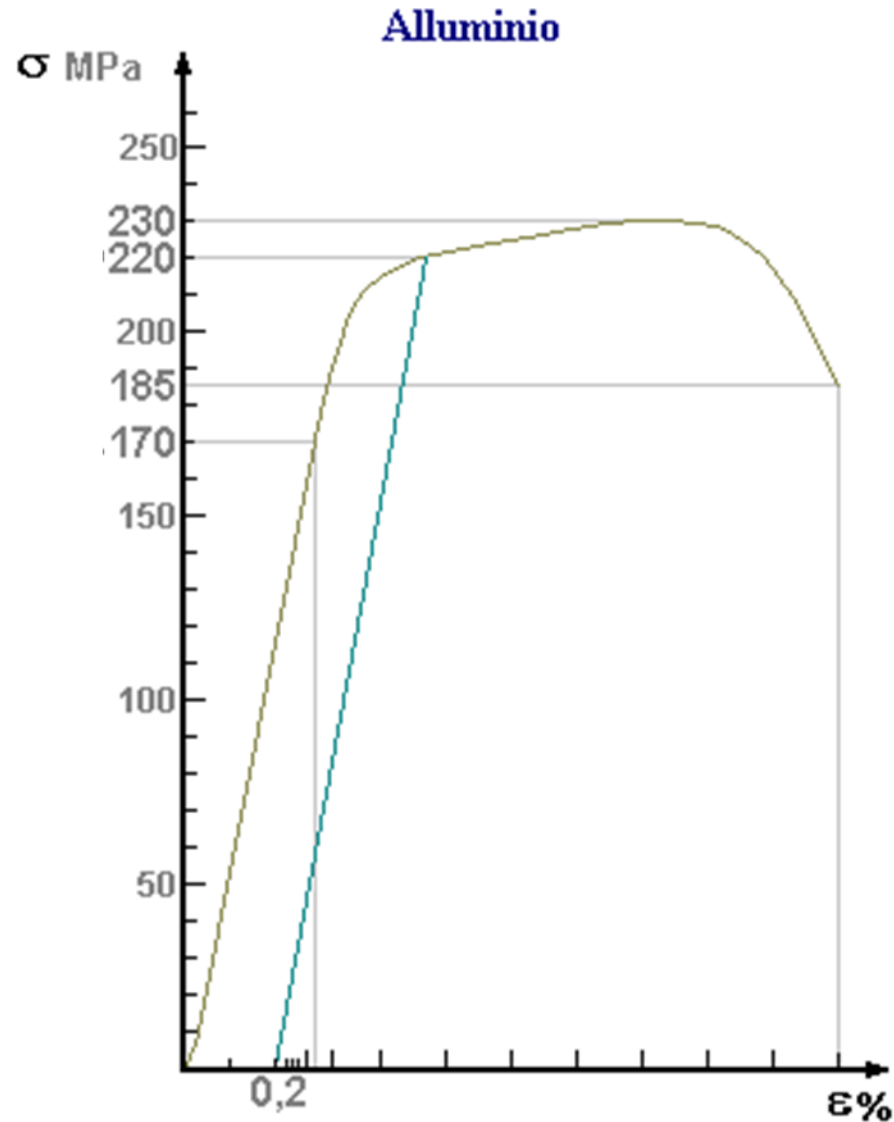
Quiz 2



The σ - ϵ curve in figure comes from the tensile test of an aluminum specimen. Select the right sentence (only one):

- a) The specimen underwent plastic deformation ✓
- b) The specimen underwent ~~fragile~~ fracture
- c) The specimen did not have permanent deformations

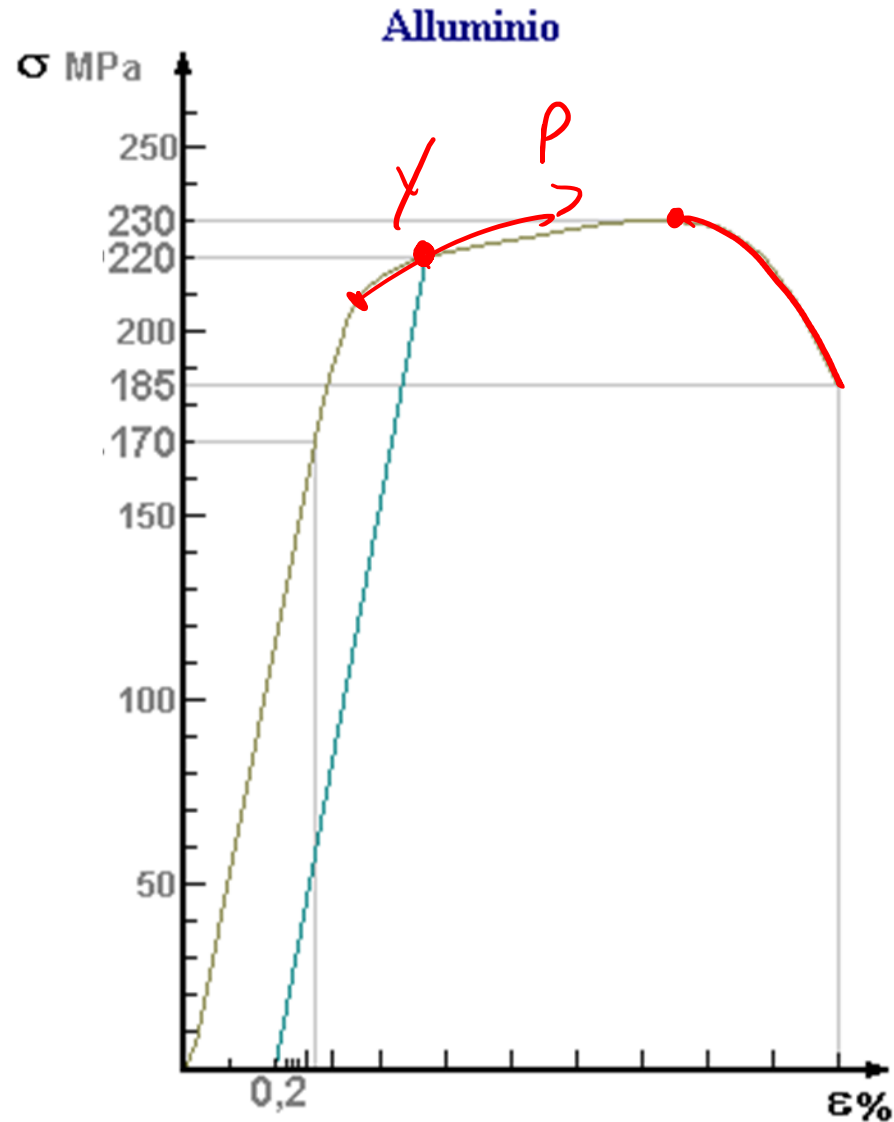
Quiz 3



The σ - ϵ curve in figure comes from the tensile test of an aluminum specimen. Select the right sentence (only one):

- a) The specimen had only elastic deformation
- b** ~~a~~) The specimen underwent necking
- c** ~~a~~) The yield strength was not reached in the test

Quiz 3



The σ - ϵ curve in figure comes from the tensile test of an aluminum specimen. Select the right sentence (only one):

- a) The specimen had only elastic deformation X
- a) **The specimen underwent necking** V
- a) The yield strenght was not reached in the test X